Background. Storytelling is a vital aspect of naturalistic language use that requires the successful integration of characters, events, and actions [1]. Successful stories include this necessary information with effective transitions between utterances that are used to convey the overall gist. The coherence of discourse is often indexed using fluency measures (interruptions in fluent speech such as um or uh) and measures of coherence (thematic organization across discourse). These are each informative in their own regard: fluency measures provide a measure of local fluency, or how fluent speech is within an utterance, while measures of coherence provide a measure of global fluency, or how thematically organized speech is overall. Recently, computational speech graph analyses have been introduced as an additional means of measuring more structural differences in discourse. Using SpeechGraph software [2,3], we quantify connectedness of an individual's discourse in a network-based approach. Since older adults have been shown to produce increasingly off-topic and locally repetitive speech [4,5], we hypothesized that discourse in older adults would become less well-connected as defined by these speech graph measures.

Methods. Across three studies, a total of 268 cognitively healthy monolingual English-speaking adults aged 20-81 (M = 46.88, SD = 17.20) completed a battery of cognitive assessments (forward and backward digit span, semantic and phonemic verbal fluency, reading span, and Stroop). All participants provided a story to the wordless picture book "*Frog, Where Are You?*" by Mercer Mayer. Additionally, participants in studies 2 & 3 completed an open-ended prompt task where they were given three minutes to respond to the question "*What do you like or dislike about living in central Pennsylvania?*". These responses were recorded and later transcribed using CLAN, a software that facilitates speech transcription and analysis [6].

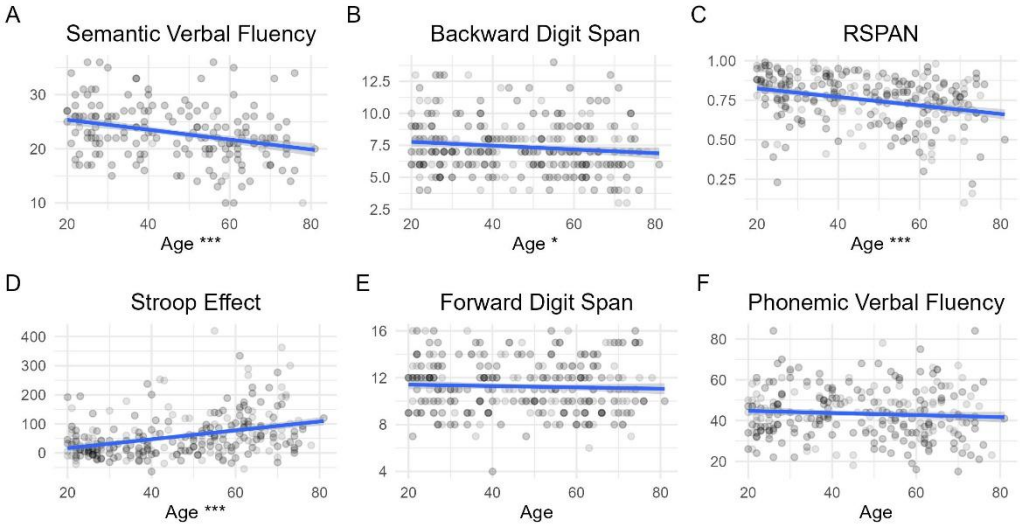
Results. These data indicated that age was associated with larger loops of 1 and 2 nodes (i.e., more local repetition), and smaller largest strongly connected (LSC) components, suggesting that older adults produce less connected speech. We also examined the effect of task and found a main effect of prompt type: showing that repeated edges, parallel edges (repetition), LSC, degree, and CC were higher in the picture book task, and LCC and ASP are higher in the open-ended prompt. Additionally, we investigated the interaction between age and prompt type for LSC indicating a negative association between LSC and age in the open-ended prompt but not in the picture book task.

Discussion. Overall, the findings suggest discourse structure changes across adulthood in nuanced ways that are not fully captured by traditional fluency or coherence measures. The patterns observed here point to a gradual shift toward speech that is more locally constrained, particularly when speakers must generate and organize content without external support. This may reflect age-related differences in how individuals manage lexical activation, maintain broader narrative threads, or draw on contextual scaffolds during extended speech. Importantly, graph-based measures appear to be sensitive to these subtle organizational shifts, highlighting their potential to complement established discourse metrics and to deepen our understanding of speech across adulthood.

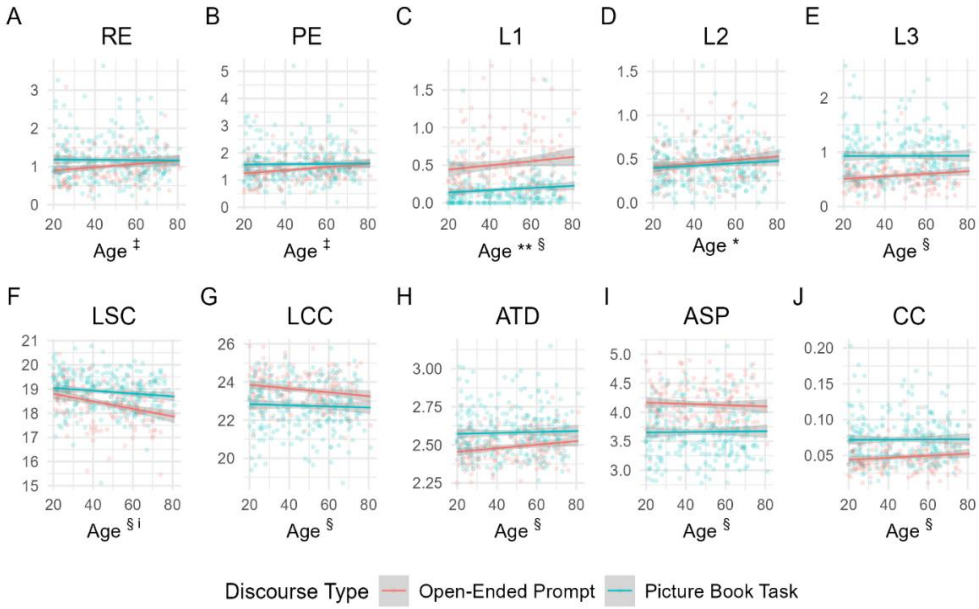
References. [1] Labov & Waletzky 1997. Journal of Narrative and Life History. [2] Mota et al., 2012. PLOS One. [3] Mota et al., 2014. Sci Rep. [4] Beier et al., 2023. Psych & Aging. [5] Glosser & Deser 1992. J Gerontol. [6] Macwhinney 2000. The CHILDES Project.

Demographics information	
N	268
Age (M [SD])	46.88 (17.20)
Gender (F/M/Non-binary)	164/103/1
Education (M [SD])	17.29 (2.79)

Correlation with Age	***β***	***t***	***p***
MoCA/MMSE	-0.20	-3.25	.001
Digit Span Backward	-0.17	-2.33	.02
Semantic Verbal Fluency	-0.30	-4.19	< .001
RSPAN (Refl. Partial Unit)	0.33	5.63	< .001
Stroop Effect	0.35	5.24	< .001



Note. * represents significant effects of age; *: p<0.05; **: p< 0.01; ***p<0.001. The non-transformed partial unit scores were used to graph RSPAN (panel A) for readability.



Note. Standard asterisks denote significant effects of age (* *p* < .05, ** *p* < .01, *** *p* < .001), while daggers († *p* < .05, ‡ *p* < .01, § *p* < .001) indicate significant main effects of task and *i* indicates a significant Age x Task interaction.